The Level 4 Natural Geometry: Two-Center Two-Electron Molecules in Molecule-Frame Hyperspherical Coordinates*

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The natural geometry hierarchy assigns each separable quantum system a coordinate system in which Coulomb singularities become boundary conditions and the problem becomes sparse and discrete: S^3 for hydrogen (Paper 7; there the discrete object is the exact label set of the S^3 harmonics, not an edge-Laplacian discretization, which does not converge), prolate spheroidal for H_2^+ (Paper 11), and hyperspherical for helium (Paper 13). The two-center, two-electron molecule H_2 combines all five Coulomb singularities and has resisted a single natural coordinate system. We introduce *molecule-frame hyperspherical coordinates* $(R_e, \alpha, \theta_1, \theta_2, \Phi)$ parameterized by the dimensionless ratio $\rho = R/(2R_e)$, where R is the internuclear distance and R_e the electronic hyperradius. The electron–electron cusp appears at $\alpha = \pi/4$, $\theta_{12} = 0$, independent of molecular geometry, where in prolate spheroidal coordinates the same locus is not a coordinate surface. That is a structural difference in where the cusp sits, and—see the scope note below—it does not translate into an accuracy advantage. The angular eigenvalue problem depends on (R, R_e) only through ρ, collapsing a two-dimensional parameter sweep to one dimension. A coupled-channel expansion in body-frame partial waves (l_1, m_1, l_2, m_2) with $l_1 + l_2$ even (gerade symmetry) and $m_1 + m_2 = 0$, solved with a variational two-dimensional (R_e, α) solver that avoids the adiabatic approximation, recovers 96.0% of the exact dissociation energy D_e at $l_{\max} = 6$ with σ and π channels and a Schwartz cusp correction (61 channels, CBS extrapolation ∼97%; the pure-variational value is ∼95–96%, this paper's two estimates of the cusp contribution being unreconciled). **[SCOPE]** This figure was previously compared against the 92.4% of the prolate spheroidal CI with algebraic Neumann integrals (Paper 12) and the difference read as an advantage of hyperspherical coordinates. That comparison was not like-for-like: Paper 12's 92.4% is a σ-only result, its basis carrying no azimuthal dependence, whereas the Level 4 figure includes π channels. Nor is a corrected comparison available in the other direction: Paper 12's own basis with $|m| \leq 1$ reaches 99.1%, but that figure and the ones here differ in truncation, cusp treatment and solver class alike, so no coordinate-system advantage is claimed in either direction. The framework extends to heteronuclear diatomics (HeH^+, $Z_A = 2$, $Z_B = 1$) via a charge-center origin shift and removal of the gerade constraint; the adiabatic solver recovers 93.1% of the exact D_e at $l_{\max} = 4$ ($\sigma+\pi$, 57 channels), but—unlike H_2—a consistent 2D variational treatment at the same truncation recovers only ∼5%, the absolute energy error (2.4%) being comparable to the small 0.075 Ha binding. The He-like ($Z = 2$) core and charge asymmetry thus converge substantially more slowly than H_2, and the adiabatic value is a non-variational overbinding. The solver has a double-adiabatic coupled-channel structure—nuclear coordinates as outer slow variable, hyperradius as inner slow variable, angular channels as fast variables—generalizing the single-adiabatic expansion of Paper 13.

I. INTRODUCTION

The natural geometry principle, articulated in Paper 13 [11], asserts that every separable quantum system possesses a natural coordinate system in which Coulomb singularities become boundary conditions rather than coordinate singularities, enabling sparse graph discretization with no fitted parameters. Three levels of this hierarchy have been established and validated:

Level 1: *One-center, one-electron* (H). Fock's 1935 stereographic projection [1] maps the hydrogen atom to the unit S^3, where the $1/r$ Coulomb singularity becomes the conformal factor of the projection. The S^3 Laplace–Beltrami operator is mathematically equivalent to the Schrödinger equation (Paper 7 [7]; 18/18 symbolic proofs), and the discrete lattice's nodes are exactly its labels (n, l, m); the lattice's graph Laplacian does not converge to it (Paper 7).

Level 2: *Two-center, one-electron* (H_2^+). In prolate spheroidal coordinates (ξ, η, ϕ), the two nuclear Coulomb singularities $1/r_A + 1/r_B$ separate into independent equations. Paper 11 [8] reaches machine precision for H_2^+ with a spectral Laguerre radial solver (1.01% with the original finite-difference lattice).

Level 3: *One-center, two-electron* (He). Hyperspherical coordinates $(R_e, \alpha, \hat{r}_1, \hat{r}_2)$ resolve the electron–electron cusp $1/r_{12}$ as a boundary condition at $\alpha = \pi/4$, $\theta_{12} = 0$. Paper 13 [11] achieves 0.05% error for He with a single-channel adiabatic lattice.

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Level 4—the two-center, two-electron molecule H_2—is the first system requiring *simultaneous* resolution of nuclear ($1/r_{iA}$, $1/r_{iB}$) and electron–electron ($1/r_{12}$) singularities. Paper 12 [10] demonstrated that prolate spheroidal CI with algebraic Neumann V_{ee} integrals saturates at 92.4% of the exact dissociation energy D_e *in a σ-only basis*. That paper originally attributed the remaining 7.6% gap to the electron–electron cusp; it has since withdrawn that attribution and identified the gap as the absent $m \neq 0$ configurations, which, restored in the same basis, take it to 99.1%.

The present paper introduces the Level 4 natural geometry: molecule-frame hyperspherical coordinates that resolve the e–e cusp while retaining the anisotropic nuclear field through a coupled partial-wave expansion. We demonstrate that this approach recovers 96.0% of D_e at $l_{\max} = 6$ with $\sigma{+}\pi$ channels (61 channels) using a variational 2D solver with Schwartz cusp correction. **[SCOPE]** Earlier versions read this as exceeding Paper 12 and confirming a cusp-resolution advantage for hyperspherical coordinates; since the Paper 12 figure it was compared against is σ-only, that inference is withdrawn. The Level 4 result stands on its own terms as a coupled-channel treatment of the two-center, two-electron problem.

II. MOLECULE-FRAME HYPERSPHERICAL COORDINATES

A. Definition

Consider two nuclei A, B with charges $Z_A = Z_B = Z$ separated by distance R along the z-axis, placed at $z = \pm R/2$. Two electrons have positions $\boldsymbol{r}_1, \boldsymbol{r}_2$ measured from the molecular midpoint. We define:

$$R_e = \sqrt{r_1^2 + r_2^2}\,, \tag{1}$$

$$\alpha = \arctan(r_2/r_1)\,, \quad \alpha \in [0, \pi/2]\,, \tag{2}$$

so that $r_1 = R_e \cos\alpha$, $r_2 = R_e \sin\alpha$. Each electron's direction in the body-fixed frame (z-axis along the internuclear axis) is specified by polar angle $\theta_i \in [0, \pi]$ and azimuthal angle ϕ_i. For Σ states (total $M = m_1 + m_2 = 0$), only the relative azimuth $\Phi = \phi_1 - \phi_2$ enters.

The full coordinate set for a Σ state at fixed R is

$$(R_e, \alpha, \theta_1, \theta_2, \Phi)\,, \tag{3}$$

with the azimuthal center-of-mass $\phi_1 + \phi_2$ integrated out.

B. The controlling parameter ρ

The dimensionless ratio

$$\rho \equiv \frac{R}{2R_e} \tag{4}$$

controls the transition between limiting geometries:

- $\rho \to 0$ (united atom, $R_e \gg R$): Level 3 He-like with $Z_{\text{eff}} = Z_A + Z_B$.
- $\rho \sim 1$ (molecular regime): full Level 4 coupling.
- $\rho \to \infty$ (separated atoms, $R_e \ll R$): two independent Level 1 atoms.

C. Volume element

The Jacobian of the transformation $(r_1, r_2) \to (R_e, \alpha)$ gives

$$dV = R_e^5 \cos^2\alpha \, \sin^2\alpha \, \sin\theta_1 \, \sin\theta_2 \, dR_e \, d\alpha \, d\theta_1 \, d\theta_2 \, d\Phi\,. \tag{5}$$

After extracting the wavefunction as $\Psi = R_e^{-5/2} (\cos\alpha \, \sin\alpha)^{-1} F(R_e) \, \Phi_\nu(\alpha, \theta_1, \theta_2, \Phi)$, the Jacobian centrifugal potential is $15/(8R_e^2)$, identical to the He case (Paper 13, Eq. 12).

D. Cusp location

The electron–electron distance is

$$r_{12}^2 = R_e^2 (1 - \sin 2\alpha \, \cos\theta_{12})\,, \tag{6}$$

so $r_{12} = 0$ requires $\alpha = \pi/4$ and $\theta_{12} = 0$. This locus is *independent of* ρ (and hence of R and R_e). The Kato cusp condition $(\partial\Psi/\partial r_{12})_{r_{12}=0} = \frac{1}{2}\Psi|_{r_{12}=0}$ is therefore a fixed boundary condition on the angular eigenfunction, unaffected by the molecular geometry.

This is the structural difference of Level 4 coordinates: the cusp is always a boundary condition rather than a coordinate surface, for *any* internuclear separation. (It is a difference in where the cusp sits; as the scope note accompanying the comparison with Paper 12 records, it does not translate into an accuracy advantage.)

III. THE MOLECULAR CHARGE FUNCTION

A. Hamiltonian decomposition

The electronic Hamiltonian at fixed R decomposes in hyperspherical coordinates as

$$H_{\text{elec}} = -\frac{1}{2}\frac{\partial^2}{\partial R_e^2} + \frac{\Lambda^2}{2R_e^2} + \frac{15}{8R_e^2} + V(R_e, \Omega; \rho)\,, \tag{7}$$

where Λ^2 is the grand angular momentum on S^5 and the potential is

$$V = V_{\text{nuc}} + V_{ee} + V_{NN}\,. \tag{8}$$

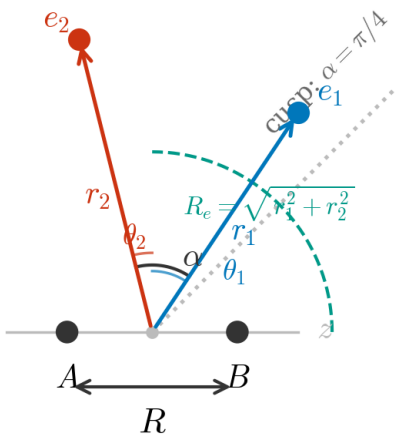


FIG. 1. Molecule-frame hyperspherical coordinates for H_2. Two nuclei A, B separated by R define the body-frame z-axis. Electrons at $\boldsymbol{r}_1, \boldsymbol{r}_2$ from the midpoint are parameterized by hyperradius $R_e = \sqrt{r_1^2 + r_2^2}$, correlation angle $\alpha = \arctan(r_2/r_1)$, and polar angles θ_1, θ_2 from the internuclear axis. The cusp locus $\alpha = \pi/4$ (dotted) is geometry-independent.

B. Electron–electron term

The e–e repulsion retains the He-like form:

$$V_{ee} = \frac{1}{R_e}\, C_{ee}(\alpha, \theta_{12})\,, \tag{9}$$

with

$$C_{ee}(\alpha, \theta_{12}) = \frac{1}{\sqrt{1 - \sin 2\alpha\, \cos\theta_{12}}}\,. \tag{10}$$

This charge function is ρ-independent: the e–e interaction depends only on the angular coordinates.

C. Nuclear attraction

The nuclear potential introduces ρ-dependence. For electron i at distance $r_i = R_e s_i$ from the midpoint (where $s_1 = \cos\alpha,\ s_2 = \sin\alpha$):

$$-\frac{Z}{r_{iA}} = -\frac{Z}{R_e}\,\frac{1}{s_i\sqrt{1 + (\rho/s_i)^2 + 2(\rho/s_i)\cos\theta_i}}\,, \tag{11}$$

with an analogous expression for nucleus B ($\cos\theta_i \to -\cos\theta_i$). The full nuclear potential is

$$V_{\rm nuc} = \frac{R_e}{R_e}\sum_{i=1}^{2}\left(-Z/r_{iA} - Z/r_{iB}\right) = \frac{1}{R_e}\, C_{\rm nuc}(\alpha, \theta_1, \theta_2; \rho)\,, \tag{12}$$

but $C_{\rm nuc}$ depends on ρ through Eq. (11), preventing a clean separation of R_e and angular coordinates.

D. The molecular charge function

Combining Eqs. (9) and (12):

$$C_{\rm mol}(\alpha, \theta_1, \theta_2, \Phi; \rho) = C_{ee}(\alpha, \theta_{12}) + C_{\rm nuc}(\alpha, \theta_1, \theta_2; \rho)\,. \tag{13}$$

In He ($\rho = 0$), the nuclear term reduces to $C_{\rm nuc} = -Z_{\rm tot}(1/\cos\alpha + 1/\sin\alpha)$ and becomes angular-only. For $\rho > 0$, the θ_i-dependence of $C_{\rm nuc}$ mixes body-frame partial waves—the source of inter-channel coupling in the multichannel expansion.

IV. DOUBLE ADIABATIC APPROXIMATION

A. Two-stage Born–Oppenheimer

The Level 4 problem has a natural double-adiabatic structure:

1. **Outer (nuclear):** Fix R; solve for the electronic energy $E_{\rm elec}(R)$.
2. **Inner (electronic):** At fixed R, decompose the electronic problem into a hyperradial coordinate R_e (slow) and angular coordinates $\Omega = (\alpha, \theta_1, \theta_2, \Phi)$ (fast).

B. Angular eigenvalue problem

At fixed R_e (and hence fixed $\rho = R/(2R_e)$), the angular eigenvalue problem is

$$\left[\tfrac{1}{2}\Lambda^2 + R_e\, C_{\rm mol}(\Omega; \rho)\right]\Phi_\nu(\Omega; R_e) = \mu_\nu(R_e)\,\Phi_\nu(\Omega; R_e)\,. \tag{14}$$

The key simplification: since $C_{\rm mol}$ depends on (R, R_e) only through $\rho = R/(2R_e)$, the angular eigenstates and eigenvalues at different (R, R_e) pairs sharing the same ρ are related by a simple R_e-rescaling. This collapses the *a priori* two-dimensional (R, R_e) sweep to a one-dimensional ρ-sweep, with substantial computational savings.

C. Liouville substitution

For the $l_1 = l_2 = 0$ channel, the Liouville substitution $u(\alpha) = \sin\alpha\,\cos\alpha\,\phi(\alpha)$ transforms the weighted Sturm–Liouville equation into standard Schrödinger form:

$$-\tfrac{1}{2}u'' + V_{\rm eff}(\alpha; \rho)\, u = \mu\, u\,, \tag{15}$$

with Dirichlet boundary conditions $u(0) = u(\pi/2) = 0$. The effective potential is

$$V_{\rm eff}(\alpha) = -2 + \frac{l_1(l_1+1)}{2\cos^2\alpha} + \frac{l_2(l_2+1)}{2\sin^2\alpha} + R_e\, C_{\rm mol}^{l_1 l_2}(\alpha; \rho)\,, \tag{16}$$

where $C_{\rm mol}^{l_1 l_2}$ is the projection of $C_{\rm mol}$ onto the (l_1, l_2) channel (Sec. V).

D. Hyperradial equation

Given $\mu_\nu(R_e;R)$, the hyperradial equation is

$$-\frac{1}{2}\frac{d^2F_\nu}{dR_e^2}+\left[\frac{\mu_\nu(R_e;R)+15/8}{R_e^2}\right]F_\nu=E_{\rm elec}\,F_\nu\,, \quad (17)$$

structurally identical to Paper 13, Eq. 18. The adiabatic potential $U(R_e;R)=[\mu(R_e;R)+15/8]/R_e^2$ is computed by sweeping R_e and solving Eq. (14) at each point.

E. Limiting cases

a. United atom ($R\to 0$). As $\rho\to 0$, the nuclear term reduces to $C_{\rm nuc}\to -(Z_A+Z_B)(1/\cos\alpha+1/\sin\alpha)$, recovering the He charge function with $Z=Z_A+Z_B$. The angular equation (14) becomes the Paper 13 equation exactly. Numerically verified: the Level 4 eigenvalue at $\rho=10^{-6}$ matches the He eigenvalue to machine precision.

b. One-electron limit. Removing electron 2 ($\alpha\to 0$, $r_2\to 0$, $r_1\to R_e$) reduces the electronic problem to the one-electron H_2^+ Hamiltonian in body-frame coordinates. The angular eigenvalue problem becomes the η-equation of prolate spheroidal coordinates (Paper 11).

V. MULTICHANNEL PARTIAL-WAVE EXPANSION

A. Channel labels

For ${}^1\Sigma_g^+$ symmetry with total $M=m_1+m_2=0$, each electron's angular part is expanded in spherical harmonics $Y_{l_i}^{m_i}(\theta_i,\phi_i)$. Channels are labeled by 4-tuples (l_1,m_1,l_2,m_2). Gerade symmetry under inversion requires

$$l_1+l_2=\text{even}\,, \quad (18)$$

and the constraint $m_1+m_2=0$ enforces Σ symmetry. Both orderings (l_1,l_2) and (l_2,l_1) with $l_1\neq l_2$ are included as separate channels because they have distinct centrifugal potentials: $l_1(l_1+1)/\cos^2\alpha$ vs. $l_2(l_2+1)/\cos^2\alpha$.

a. σ and π channels. Channels with $m_1=m_2=0$ are σ channels; the angular expansion uses Legendre polynomials $P_{l_i}(\cos\theta_i)$ and depends only on θ_1,θ_2 (no Φ-dependence). Channels with $|m_i|\geq 1$ are $\pi,\delta,\ldots$ channels; they introduce Φ-dependence through the associated Legendre functions and couple to σ channels via the e–e multipole expansion. The nuclear potential is diagonal in m (axial symmetry), so σ–π coupling arises exclusively from V_{ee}.

The channel count for $l_{\max}$ is $(l_{\max}+1)^2/2$ (approximately):

$l_{\max}$	$N_{\rm ch}$	Channels
0	1	(0,0)
1	2	(0,0), (1,1)
2	5	(0,0), (0,2), (2,0), (1,1), (2,2)
3	8	+ (1,3), (3,1), (3,3)
4	13	+ (0,4), (4,0), (2,4), (4,2), (4,4)

B. Electron–electron coupling

The multipole expansion of $1/r_{12}$ in the body-frame basis gives

$$\langle l_1'l_2'|C_{ee}|l_1l_2\rangle=\mathcal{N}\sum_k G(l_1',k,l_1)\,G(l_2',k,l_2)\,\frac{f_k(\alpha)}{\max(s_1,s_2)}\,, \quad (19)$$

where $G(l,k,l')$ is the Gaunt integral $\int_{-1}^{1}P_l\,P_k\,P_{l'}\,dx$, the function $f_k(\alpha)=[\min(\cos\alpha,\sin\alpha)/\max(\cos\alpha,\sin\alpha)]^k$, and the normalization is $\mathcal{N}=\sqrt{(2l_1'+1)(2l_2'+1)(2l_1+1)(2l_2+1)}/4$.

The selection rules on Gaunt integrals enforce sparsity: $G(l,k,l')=0$ unless $l+k+l'$ is even and $|l-l'|\leq k\leq l+l'$. In the product of two Gaunt integrals, k must simultaneously satisfy conditions from both factors.

C. Nuclear coupling

The nuclear attraction couples channels because it depends on θ_i:

$$\langle l_1'l_2'|C_{\rm nuc}|l_1l_2\rangle=N_1\int_{-1}^{1}P_{l_1'}\,V_1(x;\alpha,\rho)\,P_{l_1}\,dx\,\delta_{l_2'l_2}+\delta_{l_1'l_1}\,N_2\int_{-1}^{1}P_{l_2'}\,V_2(x;\alpha,\rho)\,P_{l_2}\,dx\,, \quad (20)$$

where $V_i(x;\alpha,\rho)=-Z/d_{iA}-Z/d_{iB}$ with $d_{iA}=\sqrt{s_i^2+\rho^2+2\rho s_i x}$, and $N_j=\sqrt{(2l_j'+1)(2l_j+1)}$ ensures consistency with the orthonormal $\hat{P}$ basis.

A single Legendre expansion of V_i in powers of ρ/s_i diverges when $\rho/s_i>1$ (electron between midpoint and nucleus). We resolve this by splitting into two convergent regions: for $\rho<s_i$, expand in $(\rho/s_i)^k$; for $\rho>s_i$, expand in $(s_i/\rho)^k$. In each region, the θ-integral reduces to a sum over Gaunt integrals

$$G(l',k,l)=2\begin{pmatrix}l' & k & l\\ 0 & 0 & 0\end{pmatrix}^2\,, \quad (21)$$

proportional to the square of a Wigner $3j$ symbol—the same objects that appear in the electron–electron coupling (Paper 13 [11], Sec. III.C). The triangle inequality $|l'-l|\leq k\leq l'+l$ forces the sum to terminate exactly at $k=l'+l$: no truncation parameter is needed. The result is exact to machine precision in all regimes, including

the boundary $\rho/s_i = 1$ where the two expansions match continuously.

Gauss–Legendre quadrature is used only in the screened ($Z_{\rm eff}$) path for composed geometries (Paper 17), where the smooth correction $[Z_A - Z_{\rm eff}(r)]/r$ is non-polynomial and the algebraic expansion does not apply.

a. Critical observation. The nuclear coupling requires *one* l-index to match via the Kronecker delta while the other index transitions via the integral. Thus the (0,0) channel couples to (0,2) through electron 2 and to (2,0) through electron 1. Omitting either ordering halves the nuclear coupling strength.

D. The gerade–quadrupole constraint

The gerade constraint $l_1 + l_2 =$ even forces the first nuclear coupling from (0,0) to involve $\Delta l = 2$ transitions:

$$(0,0) \longrightarrow (0,2),\ (2,0) \qquad \text{(quadrupolar)}\,. \tag{22}$$

Dipolar transitions ($\Delta l = 1$) would couple to (0,1) or (1,0), but $l_1 + l_2 = 1$ is odd and excluded. This explains why $l_{\max} = 1$ barely improves on $l_{\max} = 0$: the (1,1) channel has no nuclear coupling to (0,0) (both Kronecker deltas are zero) and only weak e–e coupling.

The jump at $l_{\max} = 2$ (31% → 88%) reflects the onset of nuclear coupling.

E. Heteronuclear extension

For heteronuclear diatomics ($Z_A \neq Z_B$), three modifications are required:

a. 1. Removal of the gerade constraint. The gerade symmetry $l_1 + l_2 =$ even relies on the inversion symmetry $\theta_i \to \pi - \theta_i$ exchanging identical nuclei. When $Z_A \neq Z_B$, this symmetry is absent and the channel list includes all (l_1, l_2) pairs with $l_1, l_2 \leq l_{\max}$. At $l_{\max} = 4$, $m_{\max} = 0$, this doubles the channel count from 13 (homonuclear) to 25 (heteronuclear).

b. 2. Odd-k nuclear multipole coupling. The nuclear potential becomes

$$V_{\rm nuc}^{(i)} = -\sum_{k=0}^{\infty}\Big[Z_A\, f_k(s_i,\rho_A) + Z_B\,(-1)^k\, f_k(s_i,\rho_B)\Big]\, P_k(\cos\theta_i)\,, \tag{23}$$

where the $(-1)^k$ factor encodes the opposite-pole geometry. For homonuclear systems ($Z_A = Z_B$), the odd-k coefficient $Z_A - Z_B = 0$ and only even-k terms survive. For heteronuclear systems, odd-k terms (dipole, octupole, ...) contribute with strength proportional to $Z_A - Z_B$. The k-parity of each coupling matrix element is fixed by the 3-j symbol: $\left(\begin{smallmatrix} l' & k & l \\ 0 & 0 & 0 \end{smallmatrix}\right)$ requires $l' + k + l$ even, so k has the same parity as $l' + l$.

c. 3. Charge-center origin. The coordinate origin is shifted from the geometric midpoint toward the heavier nucleus:

$$z_0 = \frac{R\,(Z_A - Z_B)}{2\,(Z_A + Z_B)}\,, \tag{24}$$

giving nuclear distances $R_A = R/2 - z_0$, $R_B = R/2 + z_0$, and dimensionless ratios $\rho_A = R_A/R_e$, $\rho_B = R_B/R_e$. For $Z_A = Z_B$, $z_0 = 0$ and the standard midpoint origin is recovered. The charge-center origin places the heavier nucleus closer to the coordinate center, improving the convergence of the partial-wave expansion by approximately one unit of $l_{\max}$.

d. Channel weight distribution. The ground-state angular eigenfunction distributes weight across channels as follows: for H_2 at $\rho = 0.7$, 77% of the norm resides in the (0,0) channel and 3 channels suffice for 90%. For a 6:1 charge ratio (modeling O–H bonds), only 26% resides in (0,0) and 8 channels are needed for 90%. The dipolar channels (0,1) + (1,0) carry 36% of the norm—more than (0,0) itself—reflecting the strong $P_1(\cos\theta)$ component of the asymmetric nuclear potential. This spreading is entirely from the nuclear coupling; the electron–electron interaction contributes no off-diagonal channel mixing ($w(0,0) = 0.9998$ with e–e coupling only). The practical implication is that asymmetric bonds require $l_{\max} \geq 4$ for qualitative accuracy, compared to $l_{\max} = 2$ for homonuclear systems.

e. Dissociation limit. For heteronuclear two-electron systems, both electrons bind to the larger-Z nucleus at dissociation. For HeH^+ ($Z_A = 2$, $Z_B = 1$): $HeH^+ \to He + H^+$, so $E_{\rm atoms} = E({\rm He}) = -2.9037$ Ha (exact nonrelativistic).

VI. RESULTS

A. Convergence with $l_{\max}$

Table I shows the convergence of $D_e = 2E({\rm H}) - E_{\rm total}$ at $R = 1.4$ bohr, computed with the variational 2D solver (Sec. VI D). The adiabatic approximation results are shown for comparison; the ∼11% gap between adiabatic and 2D results reflects the non-adiabatic kinetic energy cost ignored by the adiabatic solver.

a. Extended $l_{\max}$ convergence study. Table II (plotted in Fig. 2) extends the convergence analysis through $l_{\max} = 6$ using the 2D solver with Schwartz cusp correction (Sec. VII K). Two features are immediately apparent: (i) a pronounced even-odd staircase in the σ-only column, and (ii) a constant ∼6.6 percentage-point additive offset from π channels, independent of $l_{\max}$.

B. Comparison with Paper 12

The σ-only result (87.0%) falls below Paper 12's σ-only 92.4%—at a different truncation and in a different solver

TABLE I. Convergence with partial-wave truncation. σ channels have $m = 0$ only; $\sigma+\pi$ includes $|m| \leq 1$. The 2D solver is variational; the adiabatic solver systematically overestimates D_e by ∼11%. $D_e^{\rm exact} = 0.174475$ Ha [4].

$l_{\max}$	$m_{\max}$	$N_{\rm ch}$	2D (%)	Adiab. (%)	Gap (%)
2	0	5	79.5	90.3	10.8
2	1	9	86.2	96.8	10.6
3	0	8	80.2	91.0	10.9
3	1	18	87.1	97.7	10.7
4	0	13	87.0	98.6	11.5
4	1	29	94.1	105.2*	11.1

*Adiabatic variational violation; the 2D result is the correct variational bound.

TABLE II. Extended convergence through $l_{\max} = 6$ (2D+cusp solver). σ-only uses $m_{\max} = 0$; $\sigma+\pi$ uses $m_{\max} = 1$ with π channels frozen at their $l_{\max} = 2$ values. This column's $l_{\max} = 4$ entry (93.6%) and the 94.1% of Table III differ in *two* respects, not one: the π channels are frozen here and solved at full $l_{\max}$ there, and this table is the 2D+cusp solver while that one is the plain variational 2D solver. The two effects partly cancel, so the difference between the entries measures neither alone; the two are different calculations, not a discrepancy. δ channels ($m_{\max} = 2$) are shown at $l_{\max} = 4$ only. $\Delta_{\rm step}$ is the gain from each $l_{\max}$ increment. CBS extrapolation via $(l + 1/2)^{-2}$ fit to $l_{\max} = 4$–6.

$l_{\max}$	σ (%)	$\Delta_{\rm step}$	$\sigma+\pi$ (%)	$N_{\rm ch}$
1	37.2	− − −	43.7	2
2	79.5	+42.3	86.2	9
3	80.2	+0.66	86.8	18
4	87.0	+6.78	93.6	29
5	87.1	+0.08	93.7	42
6	88.8	+1.69	96.0	61
CBS	—		∼97	
4 (δ)	87.6	+0.65†	− − −	37

†δ-channel ($m_{\max} = 2$) gain relative to σ-only at same $l_{\max} = 4$. Cost increase: 8.7×.

class—and adding π channels pushes the Level 4 result to 94.1%.

[SCOPE] Earlier versions read that 1.7-percentage-point difference as evidence that hyperspherical coordinates resolve the cusp better than prolate spheroidal ones. The comparison does not support it, because the two entries do not contain the same physics: the 94.1% includes π channels and the 92.4% does not—Paper 12's basis is φ-independent and spans only $m_1 = m_2 = 0$. Nor does a corrected comparison run the other way, because no matched pair exists: Paper 12's own basis with $|m| \leq 1$ reaches 99.1% at $(j_{\max}, l_{\max}) = (3, 3)$, against 94.1% here at $l_{\max} = 4$ and 96.0% at $l_{\max} = 6$ with a Schwartz cusp correction and a different solver class. A grid-based prolate spheroidal calculation in the literature [9] treats two-electron H_2 accurately in these coordinates, which settles the narrower point that those coordinates are not cusp-limited without ordering the two solvers here. What the table below establishes is that π channels are worth ∼7 percentage points *in this geometry*—which is true, and is the same lesson Paper 12 learned in its own. No claim of a coordinate-system advantage survives.

TABLE III. Comparison of Level 4 with Paper 12 prolate spheroidal CI (Neumann V_{ee}) and exact results. All at $R = 1.4$ bohr. Level 4 results use the variational 2D solver.

Method	D_e (Ha)	$D_e/D_e^{\rm exact}$ (%)
Paper 12 (Neumann V_{ee}, σ-only)	0.1612	92.4
Paper 12 (Neumann V_{ee}, $\|m\| \leq 1$)	0.1729	99.1
Level 4 ($l_{\max} = 4$, σ only, 13 ch)	0.1518	87.0
Level 4 ($l_{\max} = 4$, $\sigma+\pi$, 29 ch)	0.1642	94.1
Exact [4]	0.1745	100.0

C. Grid convergence

At fixed $l_{\max} = 3$, increasing n_α from 100 to 250 and $n_{\rm quad}$ from 24 to 40 yields only a 1.3 percentage-point improvement (88.5% → 89.8%), confirming that the partial-wave truncation—not the finite-difference resolution—is the limiting factor.

D. Variational 2D solver

The adiabatic approximation (Eq. 17) neglects kinetic coupling between angular channels as R_e varies. To obtain rigorous variational bounds, we solve the full two-dimensional (R_e, α) problem directly, building the sparse Hamiltonian

$$H_{\rm 2D} = T_{R_e} \otimes I_{\rm ang} + \left[H_{\rm ang}(R_e) + \tfrac{15}{8} I\right]/R_e^2 \qquad (25)$$

in the product space of R_e grid × angular channels × α grid, and finding the lowest eigenvalue via shift-invert Lanczos [2]. The matrix dimension is $N_{R_e} \times N_{\rm ch} \times n_\alpha$; for $l_{\max} = 4$, $m_{\max} = 1$ (29 channels, $n_\alpha = 60$, $N_{R_e} = 120$), the sparse matrix has $N = 208{,}800$ rows and $\sim 6.6 \times 10^6$ nonzeros, solved in ∼2 minutes.

The 2D solver consistently gives D_e approximately 11% lower than the adiabatic solver (Table I), confirming that the adiabatic approximation systematically overestimates binding by neglecting the kinetic energy cost of angular wavefunction variation with R_e.

E. HeH^+ convergence

Table IV shows convergence for HeH^+ ($Z_A = 2$, $Z_B = 1$) at $R = 1.46$ bohr, using the charge-center origin ($z_0 = R/6 = 0.243$ bohr; Eq. 24). The exact total energy is -2.9787 Ha and the exact dissociation energy is $D_e = 0.0750$ Ha [13], measured relative to the He ground state ($E_{\rm atoms} = -2.9037$ Ha).

The system is unbound at $l_{\max} \leq 3$ (σ-only) because the total energy lies above the He dissociation limit. At

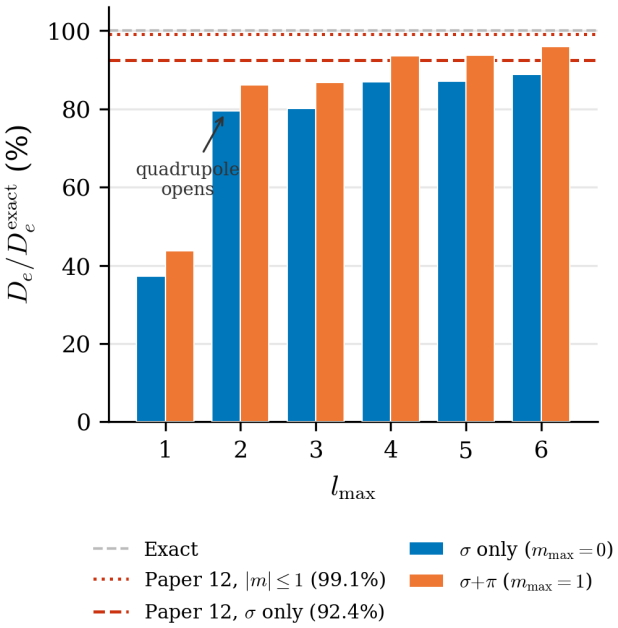


FIG. 2. Percentage of exact dissociation energy recovered as a function of $l_{\max}$, for both azimuthal sectors. Bars are the σ and $\sigma+\pi$ columns of Table II (2D solver with Schwartz cusp correction); the figure is drawn from that table and adds no data of its own. Both Paper 12 reference values are shown and each is labelled with the sector it belongs to: its σ-only 92.4% and its $|m| \leq 1$ 99.1%. Neither is a like-for-like reference for these curves—the truncation and the solver class differ—and no ordering between the two coordinate systems is asserted here or anywhere in this paper. The jump at $l_{\max} = 2$ (+42.3 points, the largest step in the table) reflects the onset of quadrupolar nuclear coupling (Sec. V).

TABLE IV. HeH^+ convergence at $R = 1.46$ bohr with charge-center origin. N_{ch} is the number of channels. The adiabatic solver is used for all entries.

$l_{\max}$	$m_{\max}$	N_{ch}	E_{total} (Ha)	D_e/D_e^{exact}
2	0	9	−2.764	unbound
3	0	16	−2.860	unbound
4	0	25	−2.961	76.8%
4	1	57	−2.974	93.1%

$l_{\max} = 4$ with σ-only channels, 76.8% of D_e is recovered; adding π channels brings this to 93.1% in the *adiabatic* treatment. This is a non-variational over-binding figure—the consistent two-dimensional variational treatment recovers only $\sim 5\%$ of D_e, so it is not parity with the variational H_2 result.

The slower convergence compared to H_2 (which is already bound at $l_{\max} = 1$) reflects the asymmetric charge distribution: even with the charge-center origin, the lighter nucleus at $\rho_B = 0.97/R_e$ requires higher partial waves to resolve. Without the origin shift ($z_0 = 0$), the system remains unbound until $l_{\max} = 4$ and recovers only

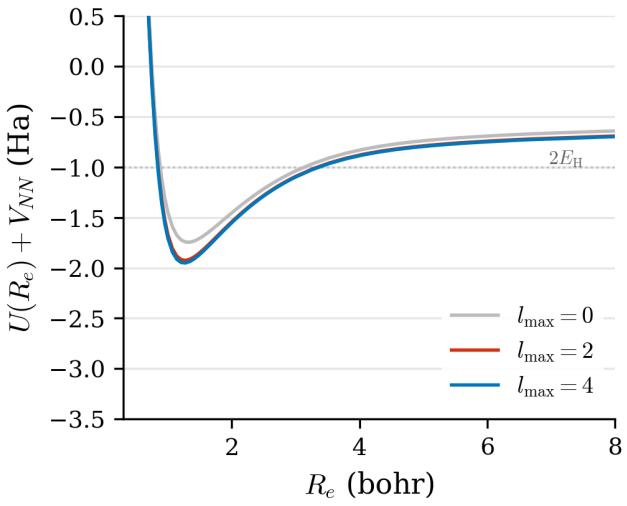


FIG. 3. Adiabatic potential $U(R_e) + V_{NN}$ at $R = 1.4$ bohr for $l_{\max} = 0$ (gray), 2 (red), and 4 (blue). The well deepens with increasing $l_{\max}$ as quadrupolar and higher nuclear coupling channels open. The dotted line marks the dissociation threshold $2E_{\text{H}} = -1$ Ha.

50.2% at $l_{\max} = 4$, $m_{\max} = 0$—the charge-center origin saves approximately one unit of $l_{\max}$.

VII. DISCUSSION

A. Why $l_{\max} = 2$ is critical

The bonding orbital σ_g of H_2 concentrates electron density along the internuclear axis. In the body-frame Legendre basis, this axial concentration requires $l \geq 1$ components of $P_l(\cos\theta)$. However, gerade symmetry forces $l_1 + l_2$ even, so the lowest channel with non-zero nuclear coupling to (0,0) is at $l_{\max} = 2$. This explains three features of the convergence:

1. The $l_{\max} = 0$ result (31%) is far below He's 99.95% at $l_{\max} = 0$—isotropic averaging destroys bonding information.

2. The $l_{\max} = 1$ improvement is minimal (31% → 37%) because (1,1) has no nuclear coupling to (0,0).

3. The $l_{\max} = 2$ jump is dramatic (37% → 88%) as the quadrupolar channels (0,2) and (2,0) open.

B. The even-odd staircase

The extended convergence data (Table II) reveals a pronounced even-odd staircase: even $l_{\max}$ increments yield large D_e gains ($l_{\max} = 2$: +42.3 pp; $l_{\max} = 4$: +6.78 pp; $l_{\max} = 6$: +1.69 pp), while odd increments

yield near-zero gains ($l_{\max} = 3$: +0.66 pp; $l_{\max} = 5$: +0.08 pp). This is not a numerical artifact but a direct consequence of the nuclear coupling selection rules.

The nuclear coupling matrix [Eq. (20)] is diagonal in one electron's angular momentum via the Kronecker delta: $\delta_{l'_2 l_2}$ for the first term and $\delta_{l'_1 l_1}$ for the second. The ground channel $(0, 0)$ therefore couples *directly* only to channels of the form $(0, L)$ or $(L, 0)$, where L is determined by the Gaunt integral selection rules. The gerade constraint $l_1 + l_2 =$ even [Eq. (18)] forces L to be even. Consequently:

- At even $l_{\max}$, new channels $(0, l_{\max})$ and $(l_{\max}, 0)$ couple *directly* to the ground channel and produce immediate D_e improvement.
- At odd $l_{\max}$, the new channels (e.g., $(1, 3)$, $(3, 1)$, $(3, 3)$ at $l_{\max} = 3$) have no direct nuclear coupling to $(0, 0)$—both Kronecker deltas are zero—and contribute only through weak electron-electron indirect coupling.

The σ-only column in Table II reproduces the staircase identically, confirming that π channels do not participate in the mechanism. A σ-only test with the e–e off-diagonal coupling removed shows that the residual odd-$l_{\max}$ gains (+0.66 pp at $l_{\max} = 3$) come from indirect coupling through the electron-electron multipole expansion, not from nuclear coupling.

C. The role of π channels

The σ-only expansion ($m_{\max} = 0$) uses Legendre polynomials and has no Φ-dependence. Adding π channels ($m_1 = \pm 1$, $m_2 = \mp 1$) introduces azimuthal flexibility and consistently improves D_e by $\sim$7 percentage points at each $l_{\max}$ (Table I). This improvement is essential. It is also, as Paper 12 has since established in prolate spheroidal coordinates, the dominant missing piece there too: the same π configurations are worth 11.6 mHa in that geometry. Comparisons between the two papers are therefore meaningful only at matched $m_{\max}$.

The π-channel coupling to σ channels arises exclusively through the e–e multipole expansion (the nuclear potential is diagonal in m by axial symmetry). The selection rule is $q = m'_1 - m_1 = -(m'_2 - m_2)$, enforced by Wigner 3-$j$ symbols in the coupling integrals.

a. The π offset. **[WITHDRAWN 2026-09-14]** [retracted 2026-09-14: p15-sigma-pi-decoupled] Earlier versions of this paragraph stated that the σ ($m = 0$) and π ($|m| = 1$) sectors are "completely decoupled in the angular eigenvalue problem", on the grounds that the nuclear coupling is diagonal in m and the e–e coupling conserves $M = m_1 + m_2$. That does not follow, and the paragraph immediately above says as much: $(m_1, m_2) = (0, 0)$ and $(+1, -1)$ both carry $M = 0$, so conserving M is precisely what places them in the *same* block, and the e–e multipole expansion couples them. The measurement points the same way—decoupled blocks would make the ground state a minimum over blocks, so adding π channels could not lower it by a roughly constant amount, which is what is observed. What survives is the empirical regularity, stated as one: π channels contribute a near-constant additive offset to D_e across σ $l_{\max}$ (Table II): the offset is 6.7 pp at $l_{\max} = 2$, 6.6 pp at $l_{\max} = 3$, 6.6 pp at $l_{\max} = 4$, 6.6 pp at $l_{\max} = 5$, and 7.2 pp at $l_{\max} = 6$. The offset is near-constant over $l_{\max} = 2$–5 and rises to 7.2 pp at $l_{\max} = 6$, the truncation that generates this paper's headline figure, so "constant" is an approximation rather than an exact regularity. The π contribution is in any case set by the frozen $l_{\max} = 2$ π channels throughout the extended study, which is a property of the methodology and not a demonstrated independence of π convergence from σ convergence.

b. δ-channel cost. At $l_{\max} = 4$, adding δ channels ($m_{\max} = 2$) yields a modest +0.65 percentage-point gain, from 87.0% to 87.6% *relative to σ-only* at the same $l_{\max}$ (the figure quoted in Table II's footnote).

[UNRECONCILED 2026-09-14] The channel count attached to this row does not fit that baseline. Table II lists $N_{\text{ch}} = 37$, and an earlier version of this sentence read that as "8 additional channels beyond the 29 σ+π channels"—but a σ+π+δ calculation at $l_{\max} = 4$ cannot return 87.6% when its own σ+π subset returns 93.6%. Either the row is σ+δ, which would be 13+8 = 21 channels rather than 37, or the 87.6% belongs to a different calculation. The two readings are not resolved here, and the row should not be relied on until they are.

The cost increase is 8.7× due to the larger matrix dimension. δ channels are therefore not cost-effective for production calculations.

D. The remaining gap and CBS extrapolation

At $l_{\max} = 6$ with σ+π channels (61 channels, $\sim$754 seconds per PES point), the 2D+cusp solver recovers 96.0% of the exact D_e. The CBS extrapolation from $l_{\max} = 4$–6 using the $(l + 1/2)^{-2}$ partial-wave convergence formula yields $\sim$97%.

The remaining $\sim$3% gap has two identified sources:

1. **Adiabatic–2D gap.** A constant $\sim$10 percentage-point gap persists between adiabatic and 2D results across all $l_{\max}$, reflecting the kinetic energy cost of non-adiabatic coupling. The 2D solver captures this correctly; the adiabatic solver overestimates D_e by this amount.

2. **Higher m channels.** δ channels ($m_{\max} = 2$) add +0.65 pp at $l_{\max} = 4$ but at 8.7× cost. **[WITHDRAWN 2026-09-14]** [retracted 2026-09-14: p15-delta-cbs-above-97] Earlier versions carried that gain into this budget and concluded that CBS extrapolation including δ "would push the limit above 97%". The +0.65 pp was measured against σ-only, not against the σ+π baseline this

budget uses, so the transfer is invalid; and the row it comes from is itself unreconciled (see the δ-channel cost paragraph above). No δ contribution to the $\sigma+\pi$ gap is claimed here.

Reaching $> 99\%$ along this route would need higher m channels ($m_{\max} \geq 2$) at extended $l_{\max}$, where the cost scaling becomes impractical. That is an extrapolation from the CBS estimate rather than a proven necessity, and it is not a necessity in general: Paper 12 reaches 99.1% at $|m| \leq 1$ in a different geometry.

a. Dissociation limit. The hyperradial solver does not converge at large R ($R_{e,\max} = 15$ bohr is too small for fully separated atoms), but this does not affect D_e because the dissociation energy is computed using the exact separated-atom energy $E_{\text{atoms}} = 2E_{\text{H}} = -1.0$ Ha rather than a large-R numerical limit.

E. Adiabatic vs. variational

The adiabatic approximation overestimates D_e by a remarkably constant $\sim$11% across all channel counts (Table I). This reflects the kinetic energy cost of angular wavefunction variation with R_e: the non-adiabatic coupling $P_{\nu\mu}(R_e) = \langle\Phi_\nu|\partial_{R_e}|\Phi_\mu\rangle$ generates off-diagonal terms in the radial kinetic energy that the adiabatic solver ignores. At moderate channel counts ($N_{\text{ch}} \leq 13$), the overcounting is masked by the larger basis truncation error. At $N_{\text{ch}} = 29$ ($l_{\max} = 4$, $m_{\max} = 1$), the adiabatic result exceeds D_e^{exact}—a variational violation that signals the breakdown of the single-channel approximation. The 2D solver (Sec. VI D) resolves this by treating R_e and the angular coordinates on equal footing.

F. Double-adiabatic coupled-channel structure

The Level 4 geometry has a double-adiabatic coupled-channel structure, extending the single-adiabatic expansion of Paper 13:

Outer (nuclear):	Slow variable = R; fast = electronic problem
Inner (electronic):	Slow variable = R_e; fast = angular channels
Coupling:	$P_{\nu\mu}(R_e) = \langle\Phi_\nu\|\partial_{R_e}\|\Phi_\mu\rangle$

The non-adiabatic coupling $P_{\nu\mu}$ encodes how the angular channel functions change as R_e varies. Since the angular problem depends on R_e only through ρ, this coupling can be computed as $P_{\nu\mu} = -(R/2R_e^2)\,\langle\Phi_\nu|\partial_\rho|\Phi_\mu\rangle$ via the Hellmann–Feynman theorem.

a. Fiber bundle interpretation. Mathematically, each adiabatic layer has the structure of a fiber bundle: the slow variable is the base space, the channel Hilbert space is the fiber, and $P_{\nu\mu}$ is the Berry connection. The Level 4 geometry thus realizes a double fiber bundle (outer: nuclear base, electronic fiber; inner: hyperradial base, angular fiber). However, the computation uses only coupled-channel methods—no connection forms or holonomy calculations are required.

G. Comparison with the He hierarchy

The He hierarchy (Paper 13) has a single adiabatic layer: hyperradius as slow variable, angular coordinates as fast. The charge function $C(\alpha, \theta_{12})$ is purely angular, so the angular eigenvalue problem at fixed R_e is a standard eigenvalue problem with no parametric dependence.

In H_2, the molecular charge function depends on ρ, introducing parametric variation of the angular problem. This is structurally analogous to the Born–Oppenheimer approximation in molecular physics: nuclear coordinates R play the role of slow parameters for the electronic problem, while R_e plays the same role within the electronic problem itself.

H. Heteronuclear systems and the charge-center origin

The HeH^+ results demonstrate that the Level 4 framework generalizes naturally to heteronuclear diatomics. The key structural change is the loss of inversion symmetry, which doubles the channel count and activates odd-k multipole coupling.

The charge-center origin (Eq. 24) is not merely a computational convenience but reflects a physical asymmetry: the electronic wavefunction is concentrated around the heavier nucleus, and placing the coordinate center there reduces the angular momentum content needed to describe the lighter-nucleus contribution. The improvement—approximately one unit of $l_{\max}$—is consistent across channel counts and suggests that the charge center is the natural expansion point for asymmetric Coulomb problems in hyperspherical coordinates.

The 93.1% recovery quoted above is an *adiabatic* value (Table IV). A consistent 2D variational treatment—the same solver that yields H_2's 94.1% at this truncation—instead gives $E_{\text{total}} = -2.9077$ Ha for HeH^+ at $l_{\max} = 4$, i.e. only $\sim$5% of D_e: the variational energy lies 2.4% (0.071 Ha) above exact, an error comparable to the entire 0.075 Ha binding. The adiabatic solver's apparent 93.1% is therefore a non-variational over-binding coincidence (the same mechanism that produces the spurious He adiabatic "0.05%" of Paper 13), not a genuine partial-wave convergence. Contrary to a naive H_2 parity, the heteronuclear $Z = 2$ system converges *substantially more slowly* than H_2: the charge asymmetry spreads angular weight across many channels (only 26% in $(0, 0)$ for a 6:1 ratio) and the He-like core demands both higher $l_{\max}$ and a finer hyperradial grid than the H_2-matched parameters used here.

a. Heteronuclear convergence and the composed-geometry boundary. For heteronuclear diatomics, the angular eigenvalue μ_0 converges exponentially with $l_{\max}$ at fixed (R, R_e), with decay rate $\beta \approx 0.42$ per unit $l_{\max}$ for a 6:1 charge ratio at $\rho = 0.7$ (exponential fit $R^2 = 0.997$). At $l_{\max} = 10$, the eigenvalue is within 1% of the Richardson-extrapolated limit.

The angular wavefunction spreading that necessitates high $l_{\max}$ is entirely nuclear in origin: removing all electron–electron off-diagonal coupling leaves $w(0,0) = 0.9998$, while removing the nuclear coupling yields $w(0,0) = 0.243$. Thus the Level 4 solver itself converges well for asymmetric bonds; the $R_{\rm eq}$ divergence reported in Paper 17 [12] is a property of the composed geometry framework (specifically, the Phillips–Kleinman pseudopotential), not of the Level 4 coordinate system.

While eigenvalue convergence at fixed geometry is exponential, the *equilibrium geometry* $R_{\rm eq}$ drifts linearly with $l_{\max}$ in the adiabatic solver: +0.262 bohr per $l_{\max}$ for bare HeH^+ (no pseudopotential, no composition). The mechanism is that asymmetric nuclear coupling populates odd-l channels—forbidden in homonuclear systems—which preferentially stabilize large-R configurations. H_2 places 94% of its channel weight in the (0,0) channel, while HeH^+ places only 68%, with 20% in odd-l channels. This drift reflects the single-channel angular truncation for the bare, asymmetric Level 4 system. It must not be conflated with the $R_{\rm eq}$ divergence of the *composed* geometry: there the variational 2D solver produces the *same* drift (Paper 17 [12]), so that divergence is structural to the PK/composed architecture, not a single-channel adiabatic artifact removable by the 2D solver.

I. Spectral Laguerre radial solver

Following the spectral Laguerre pattern established for Level 2 (Paper 11 [15]) and Level 3 (Paper 13 [16]), we replace the finite-difference R_e grid with a Laguerre spectral basis for all three Level 4 solution pathways: adiabatic, coupled-channel, and variational 2D.

The basis functions are identical to Eq. (31) of Paper 13 [16], with the substitution $R \to R_e$ and Dirichlet boundary condition enforced by a $(R_e - R_{e,\min})$ prefactor. The overlap and kinetic matrices are computed algebraically via three-term Laguerre recurrence (no quadrature), while the effective potential $U_{\rm eff}(\rho)$ is evaluated by Gauss–Laguerre quadrature at each angular sweep point ρ.

Table V shows the convergence behavior. The optimal basis size is $n_{\rm basis} = 20$; mild overlap matrix conditioning degradation appears at $n_{\rm basis} \geq 25$.

TABLE V. Convergence of the spectral Laguerre radial solver for Level 4 (H_2, $l_{\max} = 2$, $R = 1.4$ bohr). FD reference ($N_R = 400$): $E = -1.87131$ Ha.

$n_{\rm basis}$	$\Delta E_{\rm FD}$ (Ha)
10	2.5×10^{-5}
15	7.5×10^{-6}
20	5.0×10^{-6}
25	4.8×10^{-5}
30	5.8×10^{-5}

The adiabatic pathway achieves a 16× dimension reduction ($N_R = 400 \to n_{\rm basis} = 25$) with < 0.0005 Ha agreement. The 2D pathway achieves 5× reduction.

A key structural finding is that the spectral substitution does *not* improve total wall time. The angular eigenvalue sweep— building $H_{\rm ang}$ at ~ 130 ρ-values, each requiring a full angular diagonalization—accounts for 99% of Level 4 runtime. The FD radial solve uses `eigh_tridiagonal` ($O(N)$, sub-ms), which is already optimal. Accelerating Level 4 requires faster angular solves or caching strategies, not a faster radial solver.

J. Spectral Jacobi angular solver

Having identified the angular eigenvalue sweep as the 99% bottleneck (Sec. VIII), we replace the finite-difference angular solver with a spectral Jacobi polynomial basis.

The free angular Hamiltonian has exact eigenvalues given by the SO(6) Casimir: $\nu(\nu+4)/2$ for angular momentum quantum number ν. We use Jacobi polynomials as the spectral basis, with these Casimir eigenvalues as the free spectrum. The electron–electron coupling matrix V_{ee} is precomputed once via Gaunt-type angular integrals (independent of ρ), while the nuclear coupling $V_{\rm nuc}$ scales linearly with ρ. At each sweep point, the full angular Hamiltonian is assembled as

$$H_{\rm ang}(\rho) = H_{\rm free} + \rho\, V_{\rm nuc} + V_{ee} \tag{26}$$

and diagonalized.

Table VI compares the finite-difference and spectral angular solvers.

TABLE VI. Finite-difference vs. spectral Jacobi angular solver for Level 4 (H_2, $l_{\max} = 2$, 130 ρ-points).

Metric	FD ($n_\alpha = 200$)	Spectral ($n_{\rm basis} = 10$)
Matrix dimension	∼1000	50
Angular sweep time	39.9 s	0.15 s
$U_{\min}$ difference	—	$< 2 \times 10^{-5}$ Ha
D_e (% exact)	90.0%	89.7%

The spectral basis achieves a 20× dimension reduction (1000 → 50) and a 269× speedup on the angular sweep (39.9 s → 0.15 s for 130 ρ-points). The $U_{\min}$ agreement is $< 2 \times 10^{-5}$ Ha. The 0.3% D_e shift (90.0% → 89.7%) reflects removal of finite-difference error cancellation, not a regression.

Combined with the spectral Laguerre radial solver (Sec. VIII), the angular sweep fraction drops from 99% to approximately 50% of total Level 4 runtime. The angular eigenvalues $\mu(\rho)$ remain transcendental—computed by diagonalization at each ρ-point—but the per-point cost is now a 50×50 eigensolve rather than 1000×1000.

The default solver remains `angular_method='fd'`; the spectral variant is accessed via `angular_method='spectral'`.

K. Schwartz cusp correction

The electron-electron cusp at $\alpha = \pi/4$, $\theta_{12} = 0$ is resolved as a boundary condition in Level 4 coordinates, but the partial-wave expansion converges algebraically in $l_{\max}$, not exponentially. Schwartz [14] showed that the partial-wave energy increment ΔE_l for the e–e cusp converges as

$$\Delta E_l \sim -\frac{A}{(l+\frac{1}{2})^4} \tag{27}$$

where $A = (10/\pi)\langle\delta^3(\boldsymbol{r}_{12})\rangle$ depends on the coalescence density. The cumulative correction

$$\Delta E_{\text{cusp}}(l_{\max}) = -\sum_{l>l_{\max}} \frac{A}{(l+\frac{1}{2})^4} \tag{28}$$

can be applied as a post-processing correction to the variationally computed energy at any $l_{\max}$.

For He (Level 3), the cusp correction breaks through the 0.19–0.20% adiabatic floor: at $l_{\max} = 2$, the corrected error is 0.10% (from 0.24% uncorrected). At $l_{\max} = 0$, the correction overcorrects (1.6% vs 1.1% uncorrected) because the adiabatic approximation error dominates.

For H_2 (Level 4), the correction is R-dependent: largest at short R ($\sim$2.5 mHa at $R = 1.0$ bohr) and vanishing at dissociation, where the coalescence density goes to zero. The D_e differential correction is $\sim$1.7 mHa, contributing $\sim$1.0 percentage point of D_e recovery.

a. Solver independence. A comparison of the cusp correction applied to the adiabatic and 2D variational solvers at the same $l_{\max}$ confirms that the correction magnitude is solver-independent: both solvers produce identical correction values at each $l_{\max}$. The correction scales as $\sim 1/(l_{\max}+1/2)^4$, consistent with Eq. (27). At $l_{\max} = 4$, the correction differential (between R_{eq} and dissociation) is 0.39 mHa. The cusp correction is a property of the partial-wave basis truncation, not of the hyperradial solution method.

[UNRECONCILED 2026-09-14] That measurement and the $\sim$1.7 mHa / $\sim$1.0 percentage-point figure quoted above do not sit together under this paper's own $1/(l_{\max}+1/2)^4$ scaling: propagating 0.39 mHa from $l_{\max} = 4$ gives $\approx$0.09 mHa, or $\approx$0.05 pp, at $l_{\max} = 6$—not 1.0 pp. The abstract's pure-variational "$\sim$95%" is obtained by subtracting the 1.0 pp figure from 96.0%; under the scaling law it would instead be $\approx$96.0%. One of the two figures is wrong. Which one is not settled here, and the pure-variational value should be read as uncertain over $\sim$95–96% until it is.

b. Transcorrelated alternative (explored; dead end). A complementary approach attempts to remove the cusp algebraically rather than correcting it post hoc. The transcorrelated (TC) transformation $H_{\text{TC}} = e^{-J} H e^{J}$ with Jastrow factor $J = \frac{1}{2} r_{12}$ replaces $1/r_{12}$ with a smooth gradient operator. This transformation is incompatible with the adiabatic solver (the gradient operator is $O(1)$ while V_{ee} is $O(R)$ in the angular eigenvalue problem; Track BX-2). An initial second-quantized benchmark (Track BX-3) appeared to show the TC qubit Hamiltonian converging to 3.3–3.6% error for He while the standard encoding diverged (5.3% $\to$ 8.2%), but this was a *false positive*: the benchmark diagonalized the full qubit (Fock) space, including wrong-particle-number sectors, which for a non-Hermitian TC Hamiltonian admits artifactually low eigenvalues below the variational bound. Under correct particle-number-projected full CI (Track TC-V), the standard encoding does *not* diverge—it converges to $\sim$2.0%—while TC plateaus at $\sim$3.4%, i.e. *worse* than the untransformed pipeline. The cusp is therefore an energy-evaluation problem, not a wavefunction problem, and TC is not pursued further; the Schwartz post-hoc correction above remains the operative method.

c. Irreducibility of the per-ρ diagonalization. The angular eigenvalues $\mu(\rho)$ are computed by pointwise diagonalization because the nuclear coupling $V_{\text{nuc}}(\rho)$ depends on ρ through split-region factors $\min(s,\rho)/\max(s,\rho)$, which are piecewise-smooth in ρ with derivative discontinuities at $s = \rho$. Three approaches to eliminating this per-ρ diagonalization were investigated (Track Z):

1. *Algebraic blow-up*: resolving $|s-\rho|$ by introducing sheet variables produces smooth functions within each sheet, but the sheet boundaries move with ρ (at $\alpha = \arccos\rho$ and $\arcsin\rho$), creating transcendental (not polynomial) matrix elements. No global characteristic polynomial $P(\rho,\mu) = 0$ is possible.

2. *Lie algebra elevation*: the function $\min(s,\rho)/\max(s,\rho)$ is C^0 but not C^1 at $s = \rho$. Matrix elements of finite-dimensional Lie group representations are real-analytic, so this function cannot arise as a matrix element of any group containing SO(6).

3. *$S^3 \times S^3$ hybrid*: each electron's nuclear attraction is smooth on its own Fock S^3, but two-center nuclear attraction cannot be smooth on any single compact manifold (the multi-center obstruction of Papers 8–9). Furthermore, $1/r_{12}$ does not factorize on the product space.

The piecewise structure is therefore irreducible: it reflects the physical geometry of two nuclei at finite separation in hyperspherical coordinates. The spectral Jacobi solver (Table VI) is the optimal approach—it reduces the per-ρ cost from 1000×1000 to 50×50 while preserving eigenvalue accuracy.

VIII. CONCLUSION

We have introduced the Level 4 natural geometry for two-center, two-electron molecules: molecule-frame hyperspherical coordinates $(R_e, \alpha, \theta_1, \theta_2, \Phi)$ parameterized by $\rho = R/(2R_e)$.

The main results are:

1. The electron–electron cusp at $\alpha = \pi/4$, $\theta_{12} = 0$ is R-independent, resolving it as a boundary condition for all molecular geometries.

2. The angular eigenvalue problem depends on (R, R_e) only through ρ, collapsing a 2D parameter sweep to 1D.

3. The gerade–quadrupole constraint explains the dramatic jump at $l_{\max} = 2$ (31% → 80%): the first nuclear coupling from the $(0, 0)$ channel requires $\Delta l = 2$.

4. π-orbital channels ($m \neq 0$) are essential, providing ∼7 percentage points of improvement at each $l_{\max}$. The same holds in prolate spheroidal coordinates, where the corresponding configurations are worth 11.6 mHa (Paper 12); comparisons between the two geometries are meaningful only at matched $m_{\max}$, and at matched $m_{\max}$ this paper claims no advantage over Paper 12.

5. A variational 2D solver avoids the ∼11% systematic overestimate of the adiabatic approximation, giving rigorous bounds.

6. The framework extends to heteronuclear diatomics (HeH^+) by removing the gerade constraint, activating odd-k multipole coupling, and shifting the origin to the charge center (Eq. 24), recovering 93.1% of the exact D_e at $l_{\max} = 4$ with $\sigma+\pi$ channels in the *adiabatic* treatment—a non-variational overbinding figure (the consistent 2D-variational value is only ∼5% of D_e, not parity with the variational H_2 result).

The natural geometry hierarchy is now:

Level	System	Geometry	Best D_e
1	H	S^3 (Fock)	exact on the labels (diagonal in n, l)
2	H_2^+	Prolate spheroid	machine precision (spectral)
3	He	Hyperspherical	0.05% error (adiabatic FD, non-variational, error cancellation)
4	H_2	Mol.-frame hypersp.	96.0%
4	HeH^+	Mol.-frame hypersp.	93.1% (adiabatic; ∼5% variational)

The extended convergence study through $l_{\max} = 6$ with $\sigma+\pi$ channels and Schwartz cusp correction achieves 96.0% D_e (61 channels), with CBS extrapolation to ∼97%. The even-odd staircase (Sec. VII B) reflects the gerade–quadrupole constraint on nuclear coupling. π channels contribute a near-constant ∼6.6 pp offset across σ $l_{\max}$, rising to 7.2 pp at $l_{\max} = 6$. **[WITHDRAWN 2026-09-14]** [retracted 2026-09-14: p15-sigma-pi-decoupled] Earlier versions read this as consistent with complete σ–π decoupling; the two sectors are coupled through the e–e multipole expansion, and the regularity is reported here as an observation without that explanation.

a. The locked-core boundary. Extension to systems with core electrons (e.g., LiH) presents a fundamental challenge. The locked-shell approach—treating core electrons as a fixed Hartree potential while solving the active electrons in molecular coordinates—violates the discrete lattice principle of Paper 0 by converting core electrons from graph nodes to a continuous potential. This architectural mismatch produces systematic errors: the equilibrium geometry is too short ($R_{\text{eq}} \approx 2.5$ bohr vs. experimental 3.015 bohr) regardless of how the core potential is modeled (bare Z_{eff}, Clementi–Raimondi screening, or Hartree with penetration).

The resolution likely requires either (a) a full 4-electron Level 4 treatment with no locked core, which would involve an 8-dimensional angular eigenvalue problem, or (b) identification of the natural graph structure that connects atomic core geometry (S^3) to molecular valence geometry (hyperspherical), preserving discrete topology throughout. HeH^+ succeeds because it has no core electrons; the two active electrons live entirely in the molecular geometry.

b. Level 4N generalization. The Level 4 framework generalizes to $N > 2$ electrons by replacing SO(6) with SO($3N$) and the gerade constraint with full S_N antisymmetry. For LiH ($N = 4$), this "Level 4N" solver operates on S^{11} with SO(12) symmetry and S_4 singlet ($[2, 2]$) projection. The full 4-electron solver produces molecular equilibrium without a pseudopotential at $l_{\max} \geq 2$ (Paper 17, Sec. VIII), validating the composed geometry (Level 5) as an approximation to the exact N-electron treatment with 144× angular compression.

c. Note added v2.0.24. The 2D variational solver for the full 4-electron system (Track AR, v2.0.23) has confirmed that the adiabatic D_e overbinding (0.49 Ha at $l_{\max} = 2$, 5.3× the exact value) was an artifact: the 2D solver gives $D_e = -0.119$ Ha (unbound), with $E_{\min} = -7.79$ Ha (variational bound respected, above the exact -8.07 Ha). The $l_{\max} = 2$ S_4 $[2, 2]$ angular basis (12 channels) is genuinely insufficient for LiH binding. All three radial solvers—adiabatic, coupled-channel, and 2D variational—are now exhausted for the 4-electron system. The bottleneck is the angular basis, not the radial treatment. The composed geometry (Level 5) at the same $l_{\max} = 2$ achieves 5.3% R_{eq} error (vs 63.5%) because its angular pre-optimization via core/valence separation captures the dominant physics that the raw SO(12) basis cannot represent at tractable $l_{\max}$. The classical solver investigation across the natural geometry hierarchy is now complete (v2.0.24).

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